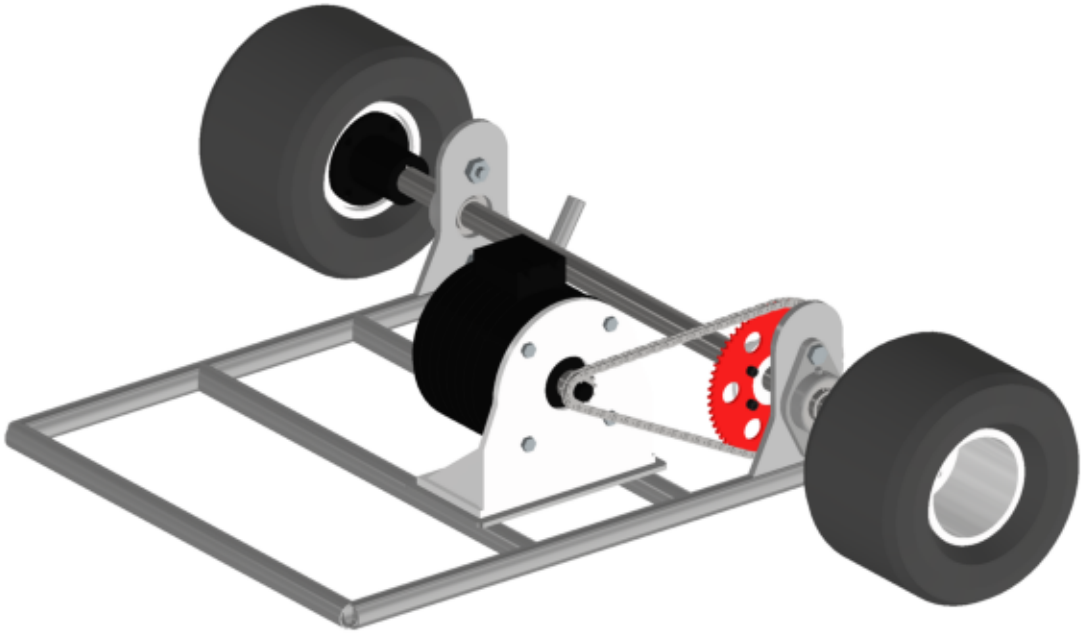
GoKartDrivetrain - calculation report

Electric go-kart rear drivetrain for 25 mph. Jay Patel, Stevens Institute of Technology (ME).

Designed and simulated only: every number is calculated or simulated from the CAD model; nothing has been built or tested.



Kart 185 kg with driver, 60 % rear, CG 280 mm, 11x7.10-5 rear tyres (r roll 137 mm)

Motor Motenergy ME1114 PMSM (0.12 Nm/A, 65 Nm peak, 5000 rpm), 350 A controller -> 41.7 Nm; 16S LiFePO4 40 Ah

Reduction ANSI 35 chain 13/60 = 4.615:1, 98 links, centres 283.9 mm; 25 mph = 3595 rpm motor

Torque at 25 mph: needs 2.56 Nm, available 17.1 Nm (margin 6.7x nominal, 2.5x low battery)

Simulated 0-25 mph 1.98 s (nominal battery; 2.41 s low), peak 0.66 g, free top speed 27.3 mph (governed at 25)

Chain tension 2095 N at full torque -> SF 4.4 on min. ultimate; speed 7.4 m/s

Axle 1-1/4 4140 PH: static SF 4.54 (nominal) / 2.69 (with keyway Kt), fatigue SF 1.97 (DE-Goodman)

Bearings UC206-20: L10 41,922 h (left, worst) at 18 mph average, static SF 2.78

Inventor: 61 occurrences, 180 constraints, 0 DOF, 0 interference; drivetrain 30.5 kg + wheels 5.4 kg

FEA Motor-mount bracket (6061-T6): worst C1 launch: 43 MPa peak / 26 MPa p99.9, SF 6.46 / 10.76

FEA Bearing hanger (4130): worst C4 kerb 2g+1g L: 129 MPa peak / 93 MPa p99.9, SF 3.57 / 4.94

FEA Rear axle (4140 PH): worst C4 kerb 2g+1g: 147 MPa peak / 139 MPa p99.9, SF 4.45 / 4.71

1 Requirements, concept and gearing

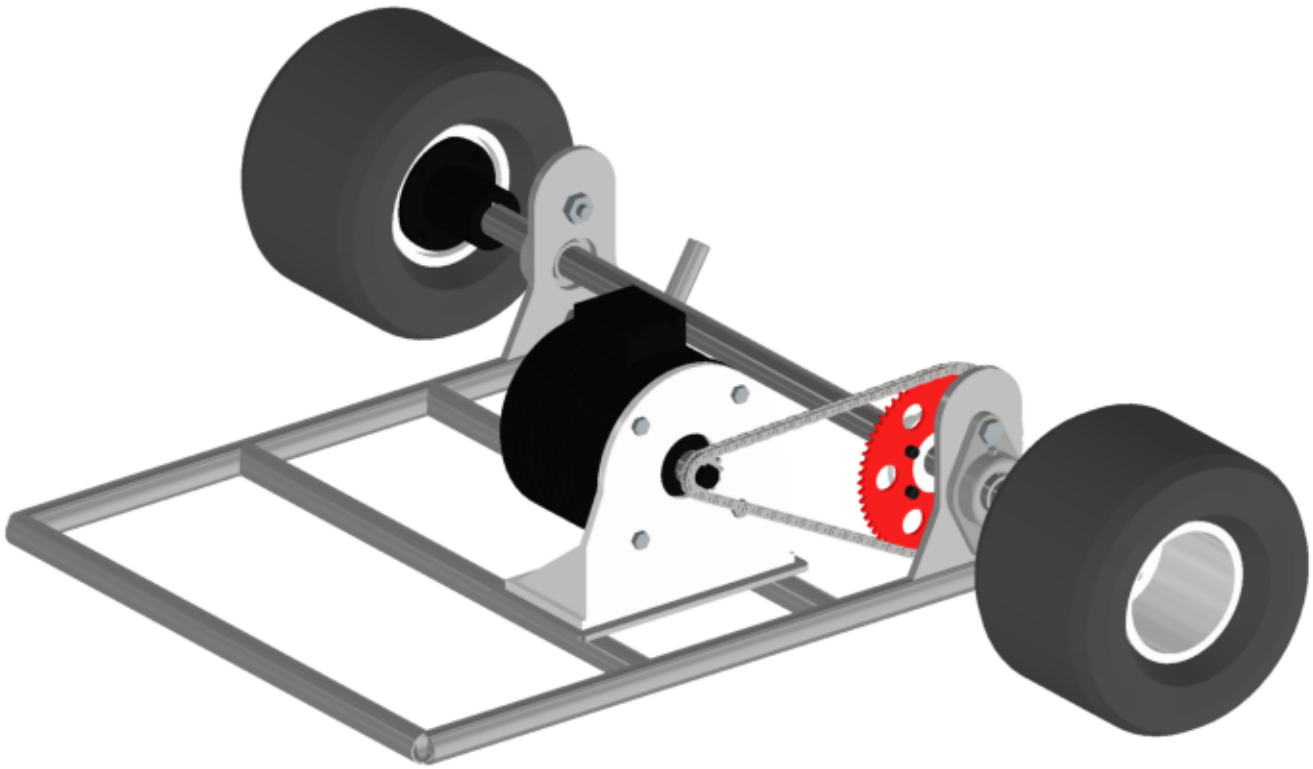
Brief: an electric go-kart drivetrain in Autodesk Inventor capable of 25 mph: electric motor, motor mounting bracket, reduction drive, rear axle, bearings, sprockets, chain and chassis mounting interfaces, with motor-torque, gear-reduction, axle-loading and acceleration calculations, FEA of the motor mount and axle supports, commercially available parts, a complete assembly, manufacturing drawings and a BOM.

Concept: a live-axle sprint/rental-class kart (11 in rear wheels) driven from the left by a single-stage ANSI 35 chain. The motor is a Motenergy ME1114 axial-gap PMSM (a common electric-kart motor) bolted by its face to a machined 6061-T6 bracket. The bracket sits in four 24 mm slots on an A36 base plate welded across two chassis cross tubes, so sliding it forward tensions the chain (assembly parameter ChainAdjust). The 1-1/4 in 4140 axle runs in two UC206-20 insert bearings in UCFL206 flange housings, bolted to 3/8 in 4130 hangers saddled onto the main rails. The 60T sprocket bolts to a keyed 6061-T6 hub. Its plane and the 13T motor sprocket are driven by the same parameter (ChainLine), so the two stay coplanar by construction.

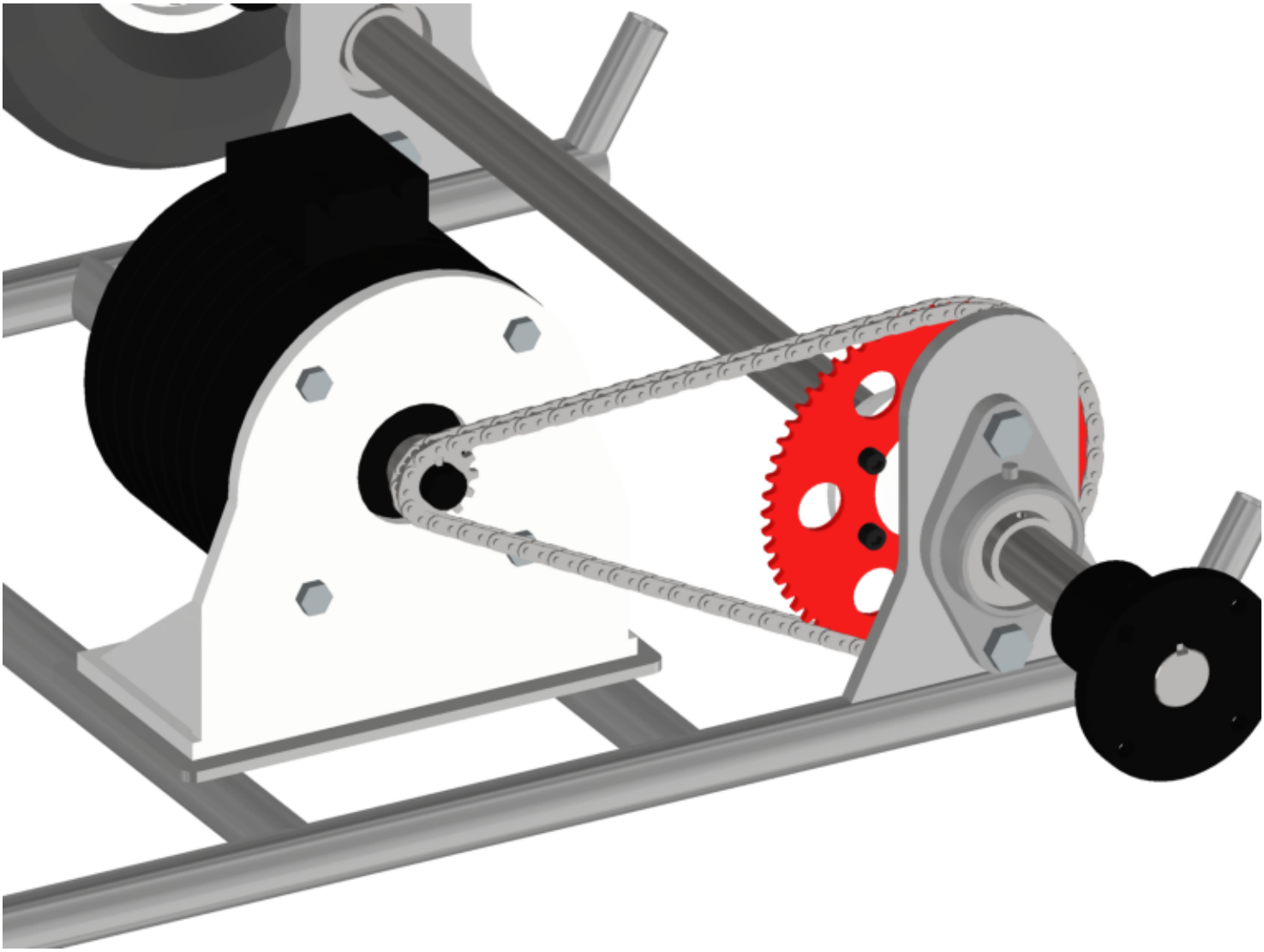
Gearing logic: top speed is set by the gearing, not by power. 25 mph at the 137 mm rolling radius is 779 rpm at the axle; the ME1114's back-EMF (Ke = Kt = 0.12 V s/rad) limits it to ~3690 rpm at a nearly flat pack (49.6 V). So the ratio was chosen to reach 25 mph even at low charge: 13/60 = 4.615 puts 25 mph at 3595 rpm (72 % of the 5000 rpm limit), and the controller's speed limit holds 25 mph on a full pack (free speed 28.1 mph). The 13T is the smallest #35 sprocket with a sound hub over a 7/8 in keyed shaft. The 60T keeps the sprocket 43 mm above the ground.

Chain geometry: the 98-link loop was closed exactly (chordal pitch-point march, closure error < 1e-12 mm), giving centres 283.889 mm. Both sprockets' tooth gaps are phased to the chain pitch points, so the modelled chain seats in the teeth (35 pitch points wrap the 60T and 6 wrap the 13T, wrap angle 151 deg).

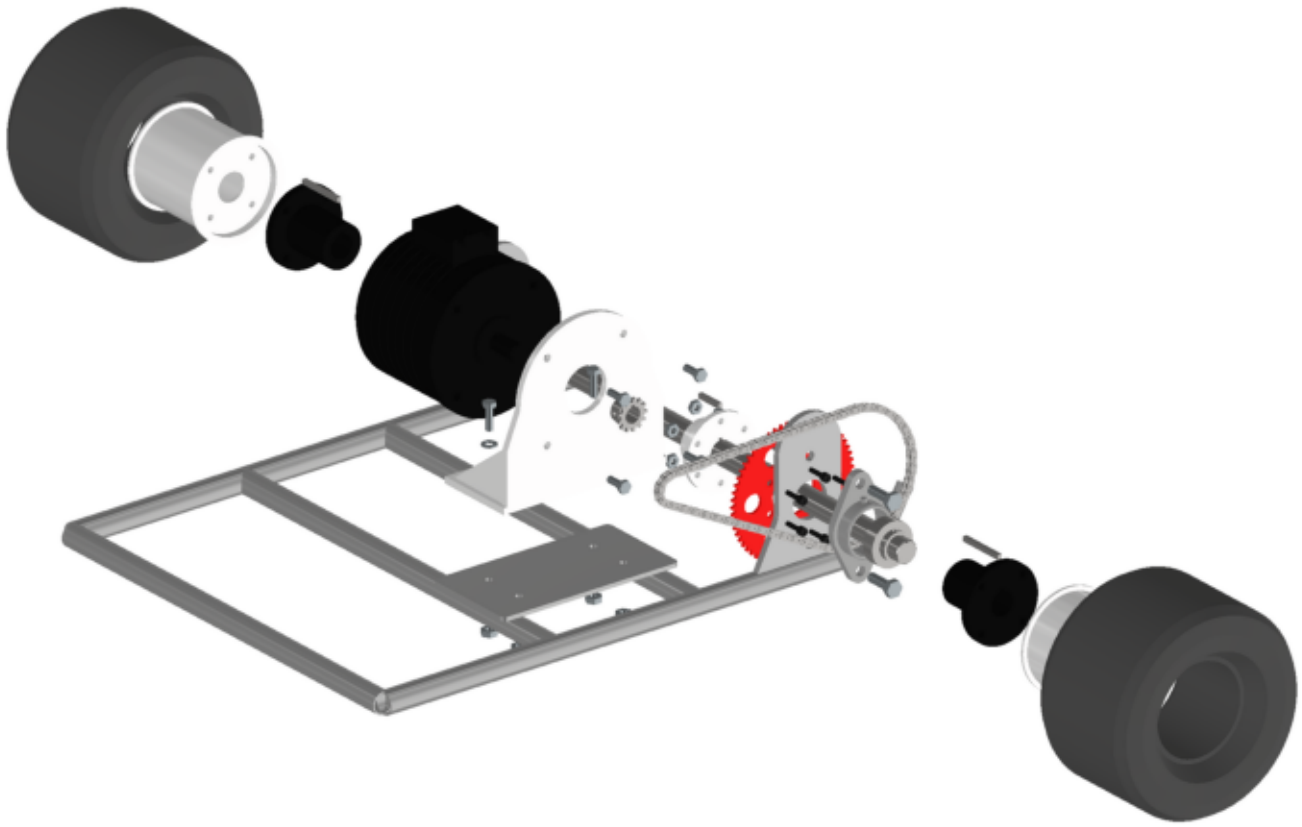
Isometric (Inventor, Realistic style)



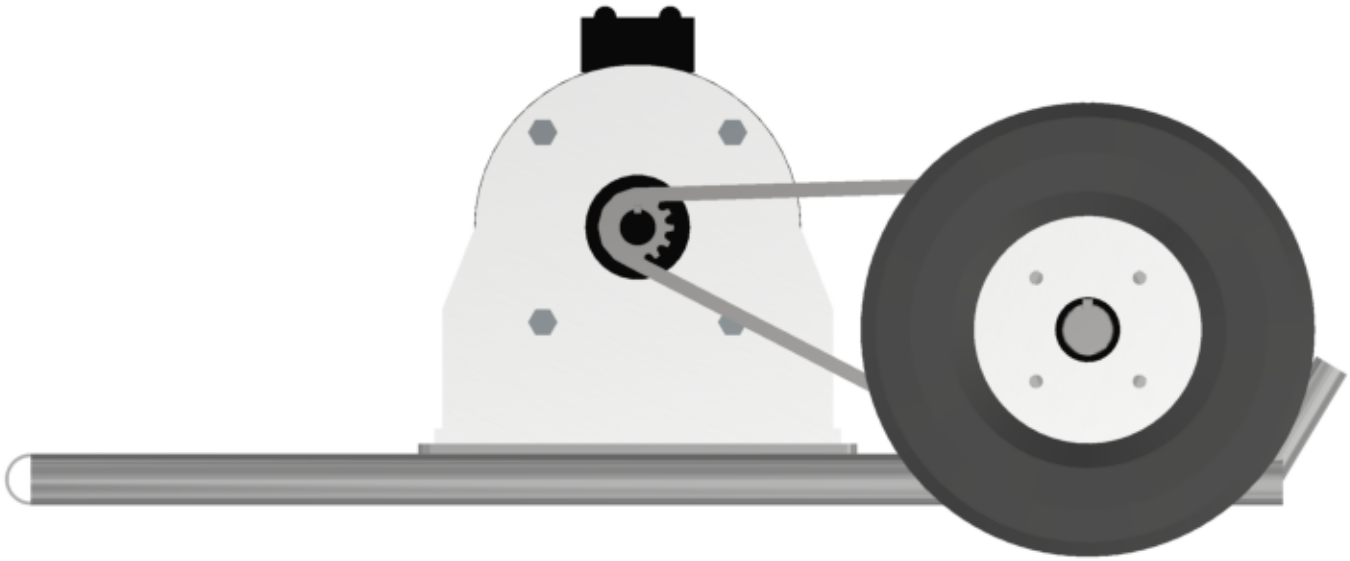
Drive close-up: 13T/60T #35 chain, slotted motor bracket, hanger



Exploded view (GoKartDrivetrain_Exploded.iam)



Left side (chain side)



2 Motor torque, road load, acceleration (simulated)

Road load at 25 mph: rolling 32.7 N (Crr 0.018) + aero 40.5 N (CdA 0.54 m2) = 73.1 N, 817 W at the wheels; referred to the motor 2.56 Nm. Cruise battery power 1080 W, so range at a steady 25 mph is ~61 km at 80 % depth of discharge (simulated).

Motor model: shaft torque = Kt I - 0.3 Nm drag. Phase current is limited by the controller (350 A), the battery (250 A battery-side) and the voltage (Ke w + R I <= 0.95 V_batt, with a 30 mOhm pack sag). The ME1114 peak stall torque (65 Nm) is above the 350 A controller limit (42 Nm), so the controller sets the launch torque.

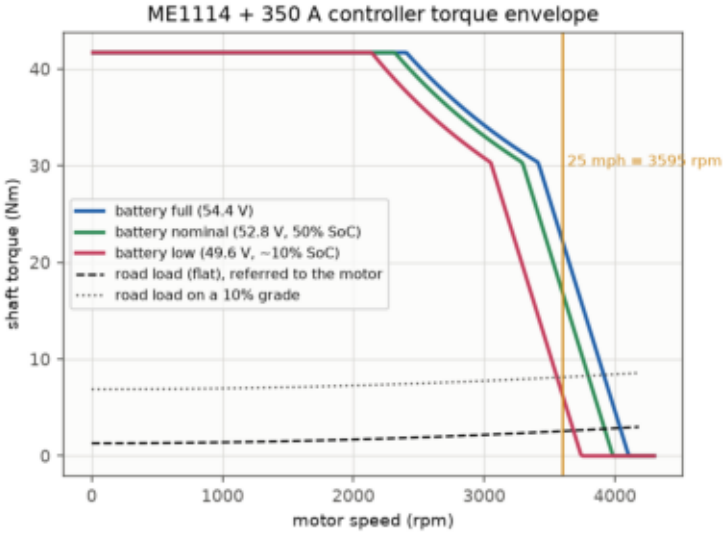
Launch simulation: explicit Euler at 1 ms. Effective mass 203.9 kg includes the wheels, axle and motor rotor (rotor 0.012 kg m2 assumed) reflected through the ratio. The traction limit is mu (1.1) times the rear load including longitudinal transfer; it never binds, because full motor torque gives 1.36 kN at the wheels against ~1.6 kN of grip. Results (s, m, g, kW):

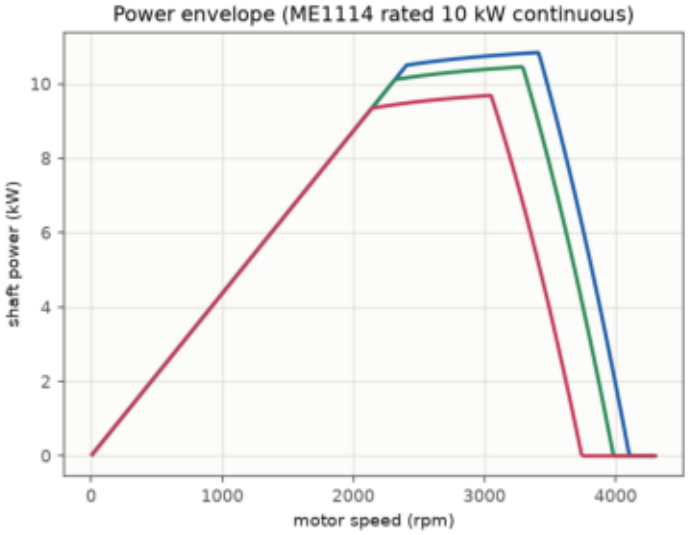
battery	T@25mph	margin	v free	0-10	0-20	0-25	d 0-25	a max	P batt
full	22.4	8.7x	28.1	0.69	1.41	1.91	11.4	0.66	11.7
nominal	17.1	6.7x	27.3	0.69	1.42	1.98	12.2	0.66	11.3
low	6.5	2.5x	25.7	0.69	1.45	2.41	16.7	0.66	10.5

Gradeability (nominal battery, steady speed the drive can hold, governor ignored): 5% 26.9 mph, 10% 26.4 mph, 15% 26.0 mph, 20% 25.6 mph, 30% 24.8 mph. Start-from-rest limit 91 % (traction-limited, theoretical).

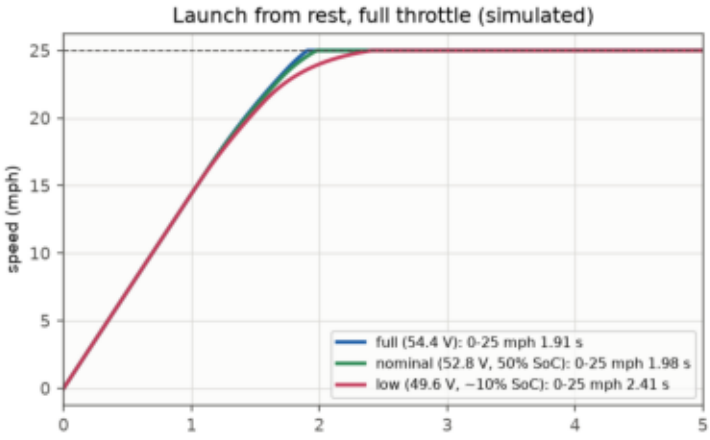
Torque margin: launch torque 41.7 Nm is 16x the 25 mph requirement; at 25 mph the voltage-limited torque available is 6.7x (nominal) and 2.5x (low pack) the road load, so 25 mph is reached at any state of charge. Cruise uses cruise 1.08 kW vs 10 kW continuous.

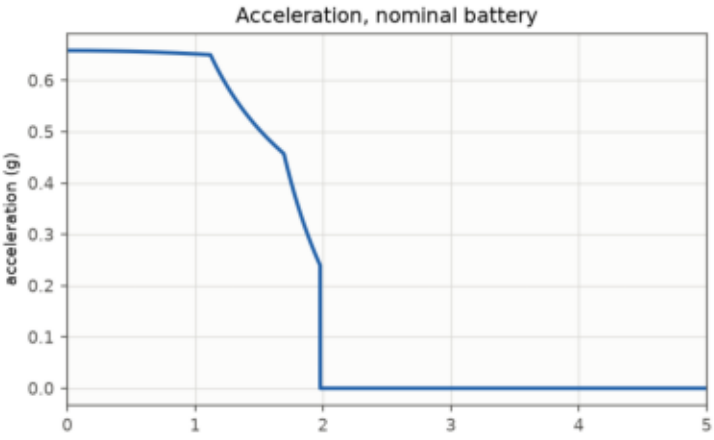
Motor + controller torque / power envelope vs road load

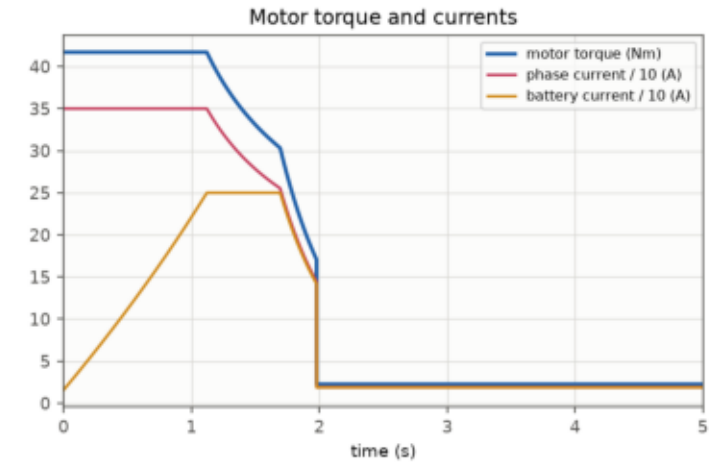


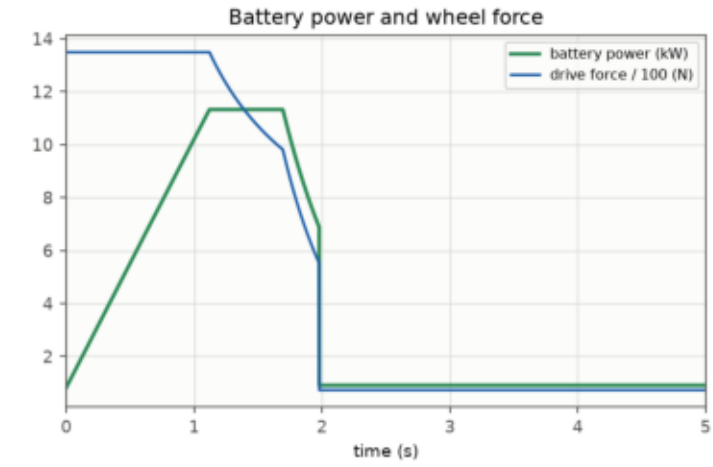


Launch simulation

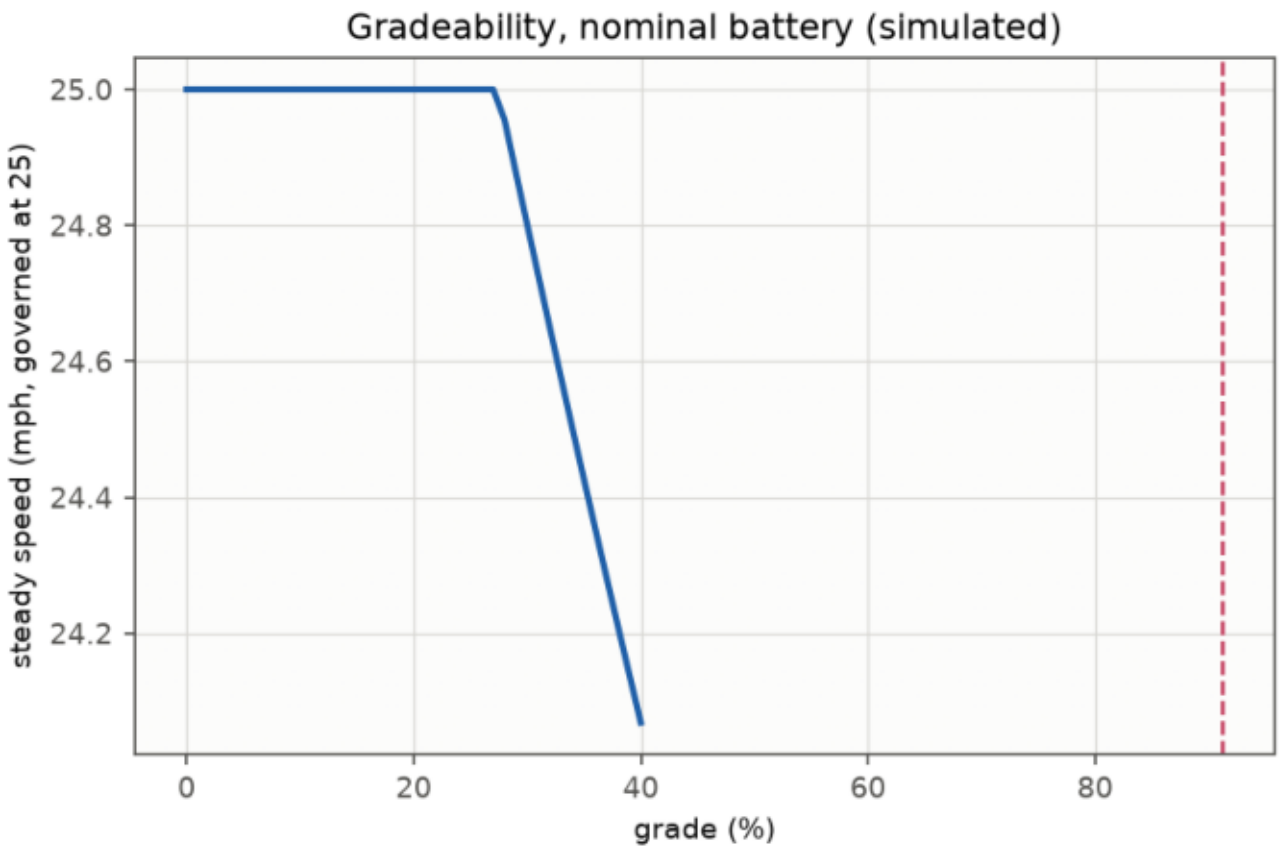








Gradeability



3 Chain, sprockets, keys

Chain: ANSI 35 roller chain (rollerless bushing chain), pitch 9.525 mm, 98 links (0.933 m), min. ultimate tensile 9.34 kN (ANSI B29.1). At 25 mph it runs at 7.42 m/s (1461 fpm), centrifugal tension 18 N.

Tight-span tension = T_motor / r_pitch(13T): 2095 N at the 42 Nm launch (SF 4.42 on min. ultimate); 2487 N if the tyres could transmit 0.75 g with mu 1.1 (SF 3.73); 129 N at 25 mph cruise (SF 64).

Power rating (Shigley eq. 17-32/33, 13T, roller/bushing-limited at 3595 rpm): 2.15 kW for 15,000 h. Cruise power x Ks 1.3 = 1.25 kW (rating SF 1.71). Full-power bursts (10.5 kW shaft) are 6.3x the 15,000 h rating: like every kart, chain wear is a maintenance item (inspect and replace at 3 % elongation). ANSI type B (bath / disc) at > 1300 fpm; on a kart: drip/spray lube each session.

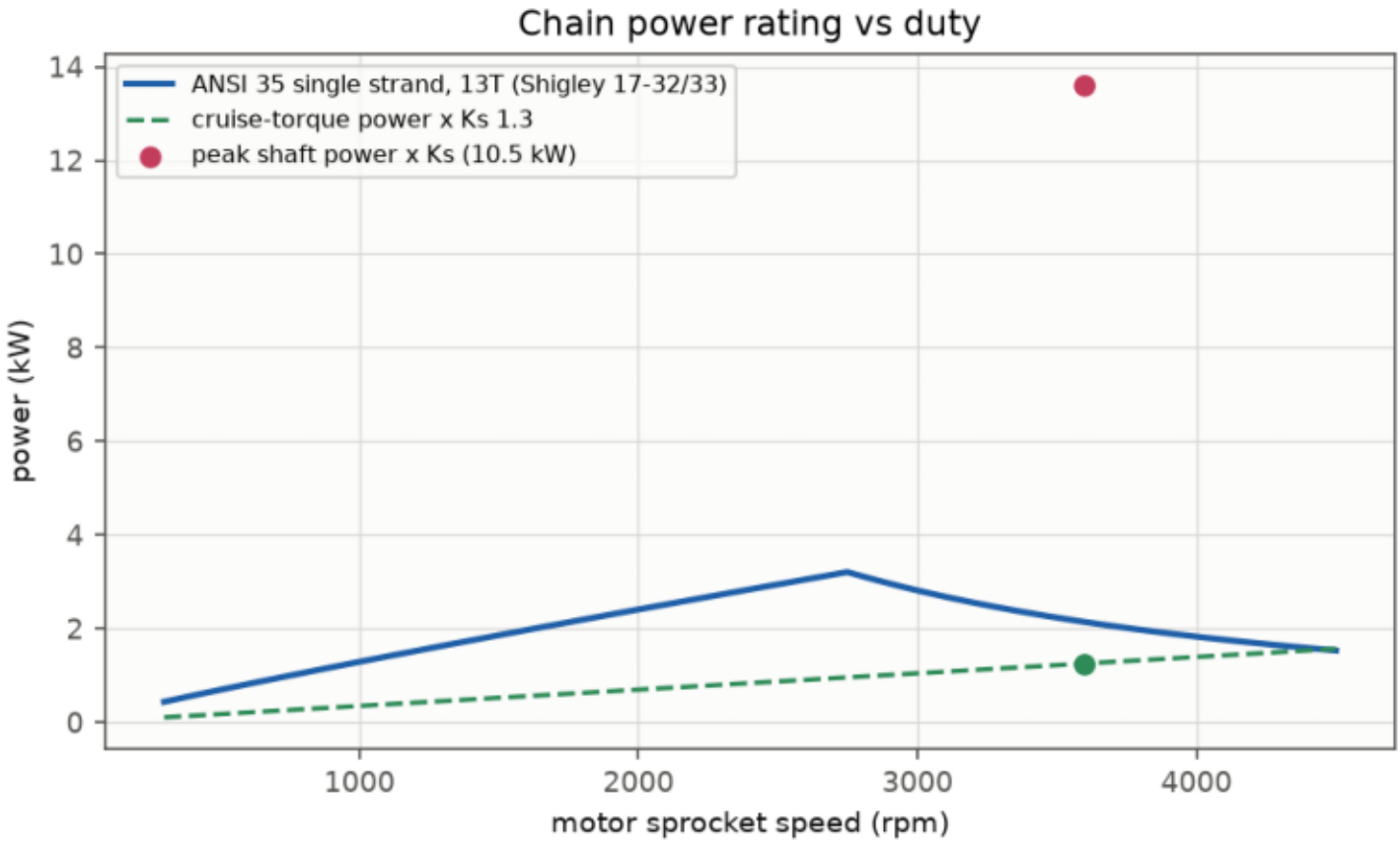
Tensioning: the 4 slots give +-12 mm, i.e. centres 272.2 to 295.6 mm (measured in Inventor), enough to take up ~2.5 % chain stretch. Slack-span sag vs centre reduction: 0 mm: 0 mm, 2 mm: 20 mm, 4 mm: 28 mm, 6 mm: 35 mm (set about 10-12 mm sag at the slack span).

Sprocket teeth (whole tension on one tooth, conservative): root shear 113 MPa -> SF 2.71 (13T 1045) / 2.57 (60T 7075-T6). Sprocket-to-hub bolts: 1111 N per 5/16 bolt, shear SF 26, bearing in the 7075 plate SF 15.

Keys (1018 CD, Sy 370 MPa; Ssy = 0.577 Sy; crushing on half the key height):

key	T Nm	shear	crush	SF shear	SF crush
axle sprocket hub key 1/4 x 1/4 x 40	189	47	94	4.56	3.95
wheel hub key 1/4 x 1/4 x 70 (whole	189	27	54	7.98	6.92
motor sprocket key 3/16 x 3/16 x 19	42	41	83	5.15	4.46

Chain power rating vs duty



4 Axle loading, bearings, mounts

Load cases (quasi-static, at the rear tyres): C1 launch: full motor torque 350 A (42 Nm), 0.75 g launch with load transfer; C2 bump 2.5g: 2.5 g vertical on both rear wheels at 25 mph cruise torque; C3 corner 1.5g: 1.5 g right turn, inner (right) rear lifted, outer (left, sprocket side) wheel carries the whole rear load and side force, 50 % torque on exit; C3R corner 1.5g: mirror of C3: left turn, outer = right wheel; C4 kerb 2g+1g: 2 x rear-axle load on the left wheel (kerb strike), 1.0 g side load, full torque.

Axle statics: wheel contact-patch forces, the chain pull on the 60T (tight upper span, toward the motor) and two bearing reactions (radial at each insert, axial shared 50/50 by the set-screw-locked inserts). Bending and torque diagrams are integrated along the axle at 1 mm. Forces N, torque / moment N m, stress MPa:

case	chain T	axle T	Fz L	Fz R	Fy L	Fy R	brg L Fr	brg R Fr	Mb max	vm	SF
C1 launch	2095	189	728	728	0	0	2618	1186	230	78	8.40
C2 bump 2.5g	129	12	1361	1361	0	0	1367	1362	313	100	6.57
C3 corner 1.5g	1048	94	1089	0	-1633	-0	2153	121	171	60	10.86
C3R corner 1.5g	1048	94	0	1089	0	1633	684	1543	171	60	10.84
C4 kerb 2g+1g	2095	189	2178	544	-1089	-0	4065	310	434	144	4.54

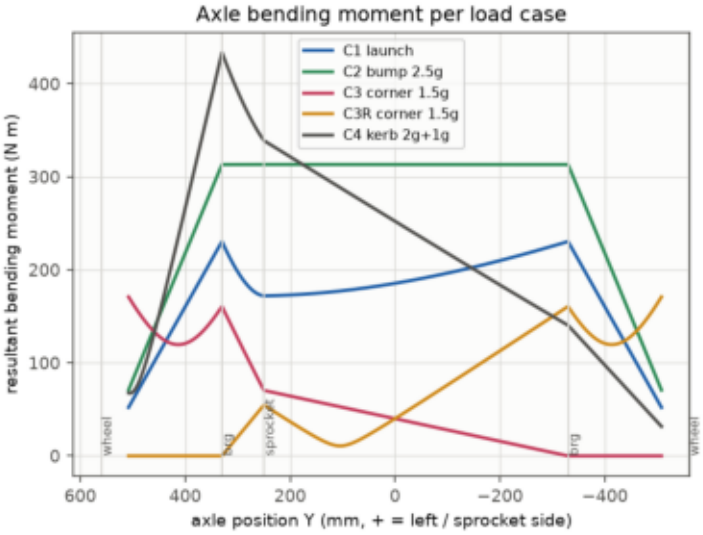
Axle stress: section modulus of the 31.75 mm bar; the full-length keyway is taken as sled-runner type (Kt 1.7, Kts 1.6, Shigley Table 7-1). Static SF on 4140 PH (Sy 655 MPa): 4.54 nominal, 2.69 with Kt (worst C4 kerb, at the bearing). Fatigue (rotating bending, steady torque, DE-Goodman, Se = 256 MPa from Marin ka 0.72, kb 0.86, 99 % reliability): SF 1.97 (C1 launch at the bearing), 2.59 for the every-corner 1.5 g case.

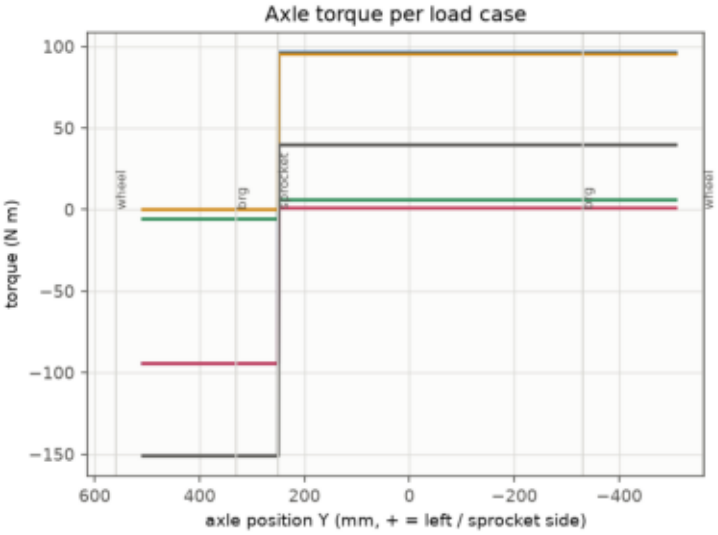
Axle deflection (simply supported at the bearings): C2 bump 2.5g: wheel 2.16 / 2.16 mm, slope 0.77 deg; C4 kerb 2g+1g: wheel 1.95 / 1.31 mm, slope 0.67 deg; C3 corner 1.5g: wheel 0.05 / 0.06 mm, slope 0.03 deg. Kart axles flex by design; UC inserts accept +-2 deg misalignment.

Bearings UC206-20 insert bearing (1-1/4 in bore) in UCFL206 2-bolt flange housing (C 19.5 kN, C0 11.3 kN). Duty cycle cruise 55%, C3 corner 1.5g 15%, C3R corner 1.5g 15%, C1 launch 12%, C2 bump 2.5g 3% at an 18 mph average (561 rpm): equivalent load 1739 N left / 1295 N right -> L10 41,922 h / 101,491 h; static SF C0/P0 2.78 / 7.32 (worst C4 kerb).

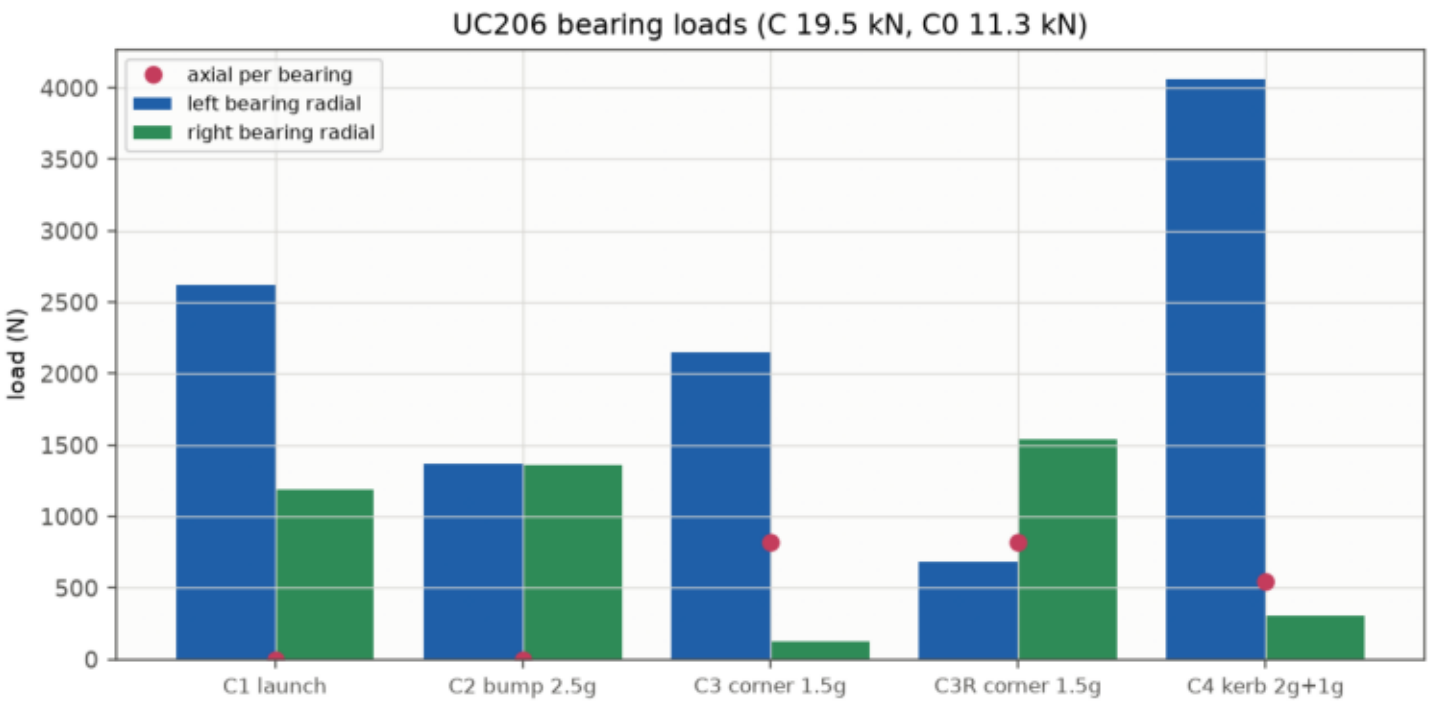
Motor mount: slot bolts 4 x 3/8-16 grade 8 at 75 % proof (31.0 kN each), friction mu 0.2 -> slip SF 11.3 against the 2191 N in-plane load, so the bracket cannot creep in its slots. Hanger welds (2 x 5 mm TIG fillet welds (ER70S-2) along the tube, 150 mm each side): 23 MPa, SF 6.2 vs the AISC 0.3 x 480 MPa allowable.

Axle bending moment and torque per load case





Bearing loads per case



5 Structural FEA (simulated)

Solver: gmsh (OpenCASCADE STEP import of the Inventor parts) + scikit-fem linear elasticity, quadratic P2 tetrahedra, PyPardiso direct solve with all load cases as one multi-RHS solve (same chain as F1Suspension / QuadDrone). An AMG-preconditioned CG fallback exists for low-RAM runs; the final results all used the direct solver.

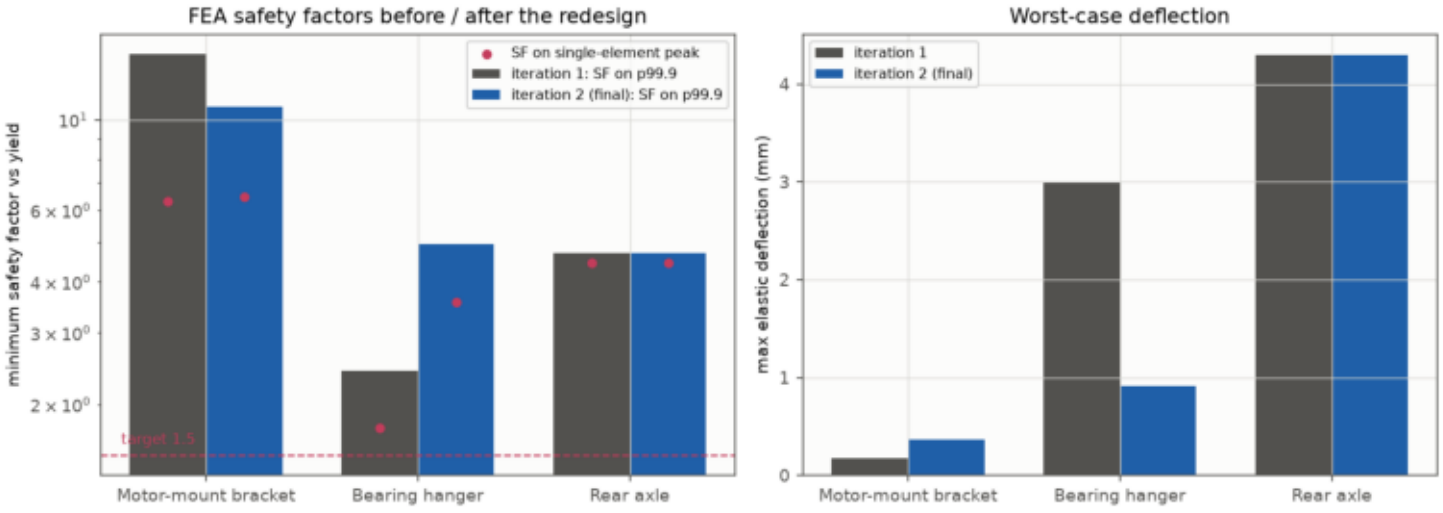
Motor bracket (6061-T6, Sy 276 MPa): the motor's in-plane load (chain pull + inertia) and its torque go through the four motor-bolt holes as cosine bearing pressure. The out-of-plane force and two bending moments act as a linear pressure on the motor-face contact ring (moments taken about the face, minus the bolt-plane offset). The four slot-bolt clamp areas under the flange are fixed. Bearing hanger (4130, Sy 460 MPa, as-welded HAZ allowable 360 MPa): the bearing force acts at the insert centre, 15 mm outboard of the plate face. Its in-plane part goes through the two 1/2 in bolt holes; the axial force and offset moments act as a linear pressure on the flange footprint. The saddle (weld to the rail) is fixed. Ten columns: the left-hanger loads of all five cases, and the right-hanger loads mirrored onto the same part, with 100 % of the axial load on the analysed bearing. Axle (4140 PH, Sy 655 MPa): wheel forces as force + bending couple + torque on the hub seats, chain load on the sprocket-hub seat, bearing reactions on the insert seats. The set is exactly self-equilibrated except the 1 % bearing-friction torque, so a 3-2-1 support carries <= 153 N. Torque on the keyway-interrupted seats is applied as force-free shear (the keys are not modelled).

Design iteration driven by the FEA (iteration 1 -> 2): the 1/4 in hanger reached 262 MPa peak / 190 MPa p99.9 (SF 1.75 / 2.42) at the re-entrant taper kink. It deflected 3.0 mm out of plane under the 2 g kerb + 1 g side case, too flexible for the bearing line. It became 3/8 in with R25 blends at the kinks: 129 / 93 MPa (SF 3.57 / 4.94), deflection 0.92 mm, +0.43 kg each. The 1/2 in motor bracket was far oversized (p99.9 SF 14.5) and became 3/8 in: 2.51 -> 1.96 kg, still p99.9 SF 10.8.

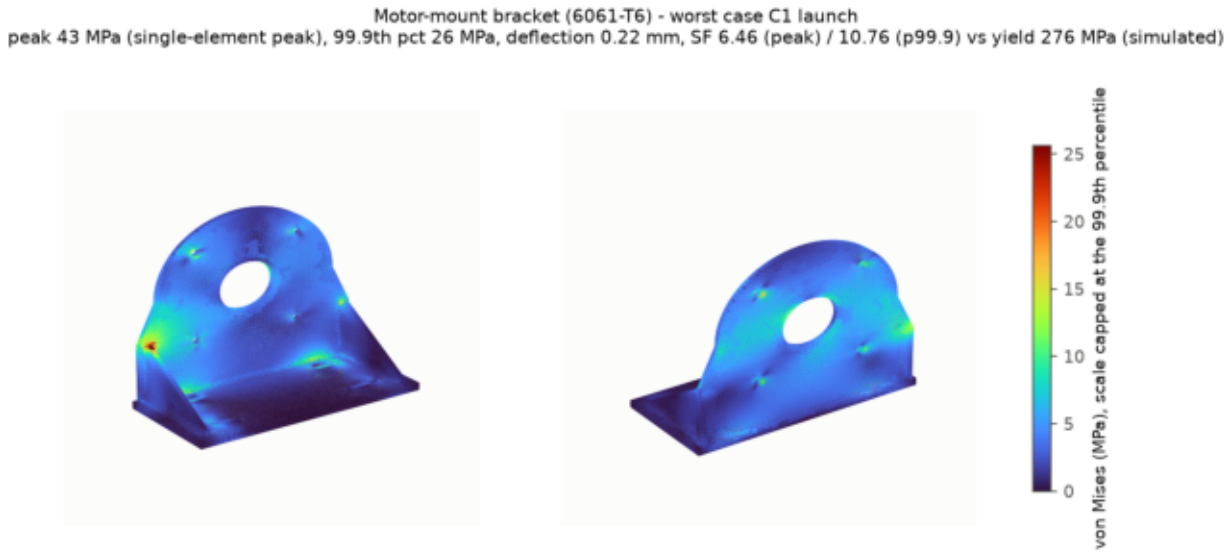
Peak vs design value: sharp CAD edges (slots, bores, weld toes) give single-element singularities. Both the peak and the 99.9th-percentile element stress are reported, and both meet the 1.5 target on yield at limit load.

part	case	DOF	peak	p99.9	u	SF peak	SF p99.9
MotorMountBrac	C1 launch	275,640	43	26	0.22	6.46	10.76
MotorMountBrac	C2 bump 2.5g	275,640	25	12	0.37	10.83	23.79
MotorMountBrac	C3 corner 1.5g	275,640	22	13	0.08	12.28	20.76
MotorMountBrac	C3R corner 1.5g	275,640	38	18	0.35	7.30	15.10
MotorMountBrac	C4 kerb 2g+1g	275,640	42	25	0.26	6.62	11.12
BearingHanger	C1 launch L	260,616	40	33	0.16	11.51	14.10
BearingHanger	C1 launch R	260,616	21	17	0.13	21.40	26.39
BearingHanger	C2 bump 2.5g L	260,616	28	23	0.23	16.17	19.63
BearingHanger	C2 bump 2.5g R	260,616	28	24	0.23	16.40	19.36
BearingHanger	C3 corner 1.5g L	260,616	119	85	0.91	3.85	5.43
BearingHanger	C3 corner 1.5g R	260,616	86	63	0.73	5.38	7.33
BearingHanger	C3R corner 1.5g L	260,616	88	65	0.73	5.26	7.05
BearingHanger	C3R corner 1.5g R	260,616	114	83	0.91	4.05	5.56
BearingHanger	C4 kerb 2g+1g L	260,616	129	93	0.92	3.57	4.94
BearingHanger	C4 kerb 2g+1g R	260,616	54	39	0.45	8.53	11.87
RearAxle	C1 launch	411,261	80	77	2.36	8.22	8.49
RearAxle	C2 bump 2.5g	411,261	108	107	4.29	6.09	6.14
RearAxle	C3 corner 1.5g	411,261	58	55	0.63	11.31	11.98
RearAxle	C3R corner 1.5g	411,261	58	56	0.78	11.28	11.79
RearAxle	C4 kerb 2g+1g	411,261	147	139	3.63	4.45	4.71

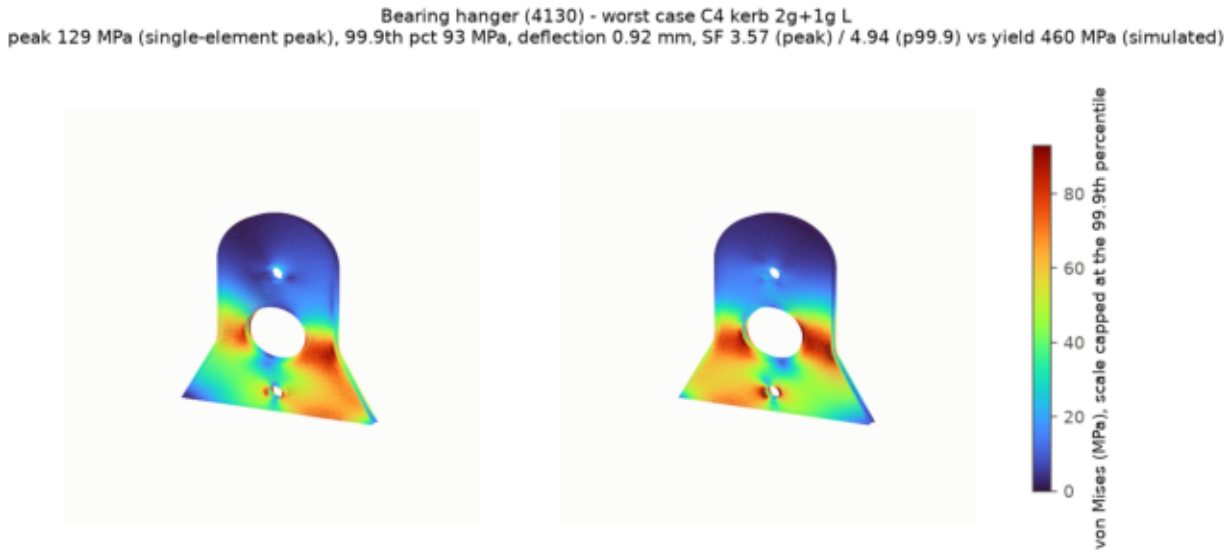
FEA iteration: before / after



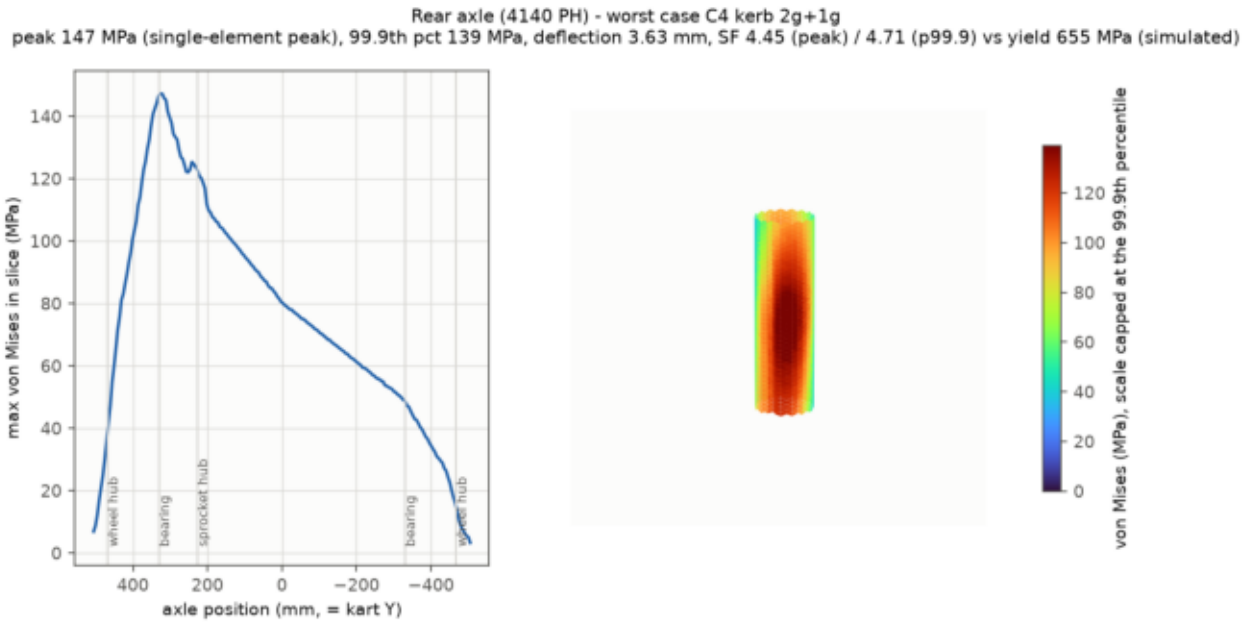
FEA - Motor-mount bracket (6061-T6)



FEA - Bearing hanger (4130)



FEA - Rear axle (4140 PH)



6 Assembly checks (Inventor)

Inventor checks: 61 occurrences (26 files), 180 constraints {'Flush': 140, 'Mate': 40}, unhealthy 0, DOF histogram {'0T/0R': 61} (total 0). Parameters {'ChainLine': '250.0 mm', 'ChainAdjust': '0.0 mm'}.

Interference (AnalyzeInterference on all occurrences, > 0.01 mm3): design pose 0 pairs; ChainAdjust -12/-6/+6/+12 mm (chain excluded, since it is modelled tight at 0): 0, 0, 0, 0 pairs; ChainLine 246 / 254 mm (chain included): 0, 0 pairs.

Sprocket alignment: over ChainLine 246-254 mm the solved 13T and 60T mid-planes and the chain plane coincide to 0.0e+00 mm, with axes parallel to 0.0e+00 deg (target < 0.5 mm / 0.1 deg).

The checks found and drove one fix: the reference chassis' rear bumper stays clipped both hanger plates (138 mm3 each) and, at ChainLine 254, the chain; the stays were rerouted.

7 Mass and BOM (Inventor)

Inventor mass properties: drivetrain 30.48 kg (CG [157.8, 92.6, 162.0]), rear wheels 5.40 kg. Motor, chain, rims and tyres carry datasheet / catalogue mass overrides.

item	qty	part no.	description	kg
1	2	GK-101	Rear-axle bearing hanger, 3/8 in 4130 plate, R25 ble	2.56
2	1	GK-102	Motor-mount base plate 270 x 142 x 6, A36, welded ac	1.79
3	1	GK-103	Motor mount bracket, 6061-T6 machined from solid: 3/	1.96
4	1	GK-301	Rear live axle 1-1/4 in x 40 in, 4140 pre-hard 28-32	6.16
5	1	GK-302	Axle sprocket hub 1-1/4 bore, 1/4 keyseat, flange 10	0.35
6	1	GK-201	Motenergy ME1114 brushless PMSM, 24-72 V, 0.12 Nm/A,	13.15
7	1	GK-401	Sprocket ANSI 35 13T, 7/8 bore, 3/16 key, B-hub (PD	0.09
8	1	GK-402	Kart sprocket ANSI 35 60T plate, 7075-T6 hard anodis	0.23
9	1	GK-403	Roller chain ANSI 35 single strand, 98 links incl. c	0.31
10	2	GK-411	Rear wheel hub 1-1/4 bore x 80 long, 1/4 keyway, 4-b	1.22
11	2	GK-421	Flange housing UCFL206 2-bolt, cast iron, bolt centr	1.04
12	2	GK-422	Insert ball bearing UC206-20, 1-1/4 bore, set-screw	0.79
13	1	GK-511	Square key 1/4 x 1/4 x 40, 1018 CD (sprocket hub)	0.01
14	2	GK-512	Square key 1/4 x 1/4 x 70, 1018 CD (wheel hubs)	0.04
15	1	GK-513	Square key 3/16 x 3/16 x 19, 1018 CD (motor sprocket	0.00
16	2	GK-412	Kart rim 5 in dia x 5 in wide, cast aluminium (envel	2.00
17	2	GK-413	Kart slick 11 x 7.10-5 (rear), envelope	3.40
18	4	GK-521	Hex head cap screw 3/8-16 x 1-1/4, SAE J429 grade 8,	0.11
19	4	GK-522	Hex head cap screw 3/8-16 x 1, SAE grade 8 (motor fa	0.10
20	4	GK-523	Hex head cap screw 1/2-13 x 1-1/2, SAE grade 8 (bear	0.25
21	6	GK-524	Socket head cap screw 5/16-18 x 1, ASTM A574 black o	0.07
22	4	GK-525	Nylon-insert locknut 3/8-16, grade 8	0.06
23	4	GK-526	Nylon-insert locknut 1/2-13, grade 8	0.14
24	6	GK-527	Nylon-insert locknut 5/16-18, grade 8	0.05
25	4	GK-528	Flat washer DIN 125 M10 (10.5 x 20 x 2), used under	0.02
	60	TOTAL (CAD, Inventor)		35.88
		of which designed (welded		12.81
		of which rear rims + tyres		5.40
		NOT MODELLED - electrical		
	1	GK-601	Motor controller, 48 V sine-wave PMSM, sin/cos input	2.90
	1	GK-602	16S LiFePO4 40 Ah pack (51.2 V nominal, 2.05 kWh), w	20.00
	1	GK-603	Main contactor 48 V 200 A + 250 A fuse + key switch	0.60
	1	GK-604	Hall throttle pedal 0-5 V	0.45
	1	GK-605	Phase and battery cable 2 AWG, lugs (estimate)	1.20
		TOTAL incl. electrical sys		61.03

8 Assumptions and limitations

Designed and simulated only. No part of this drivetrain has been built or tested; every number is a model prediction.

Vehicle: 185 kg with an 80 kg driver, 60 % rear, CG 280 mm, wheelbase 1040 mm, Crr 0.018, CdA 0.54 m2, mu 1.1 (assumed, typical kart values). Rolling radius 137 mm (11.0 in tyre, 2.7 mm deflection assumed).

Motor data from the Motenergy ME1114 listing (Thunderstruck EV): 24-72 V, 200 A cont. / 420 A peak, 65 Nm peak stall, 0.12 Nm/A, 5000 rpm, 6.5 mOhm, 92 % peak efficiency, 29 lb, 7/8 in shaft with 3/16 key. NOT on the listing, so assumed: rotor inertia 0.012 kg m2, the face bolt pattern (4 x 3/8-16 on 165.1 mm) and the motor's outer shape (envelope model). The DC-equivalent Ke = Kt model ignores field weakening and iron loss (0.3 Nm drag lumped).

Controller (350 A phase / 250 A battery, Sevcon Gen4 Size 4 class) and battery (16S LiFePO4 40 Ah, 30 mOhm) are class assumptions; neither is modelled in CAD (listed separately in the BOM with estimated masses).

Purchased parts without vendor CAD (sprockets, chain, UC206 / UCFL206, hubs, rims, tyres, keys) are modelled from catalogue dimensions (tooth form: ANSI 35 seating curve with straight 20 deg flanks); ratings: chain 9.34 kN min. ultimate (ANSI), UC206 C 19.5 / C0 11.3 kN. Fasteners and washers are step.parts STEP files (DIN 125 M10 washer used as the 3/8 washer).

Loads are quasi-static limit cases (no dynamic amplification measured). The FEA is linear elastic with nominal geometry, no weld fillets and no keys; the hanger weld zone is checked against an assumed 360 MPa as-welded 4130 allowable. Fatigue is checked analytically for the axle only.

The chassis section is reference geometry. The chain is modelled tight at ChainAdjust = 0. Seat, steering, brakes, chain guard and the electrical installation are not modelled.